

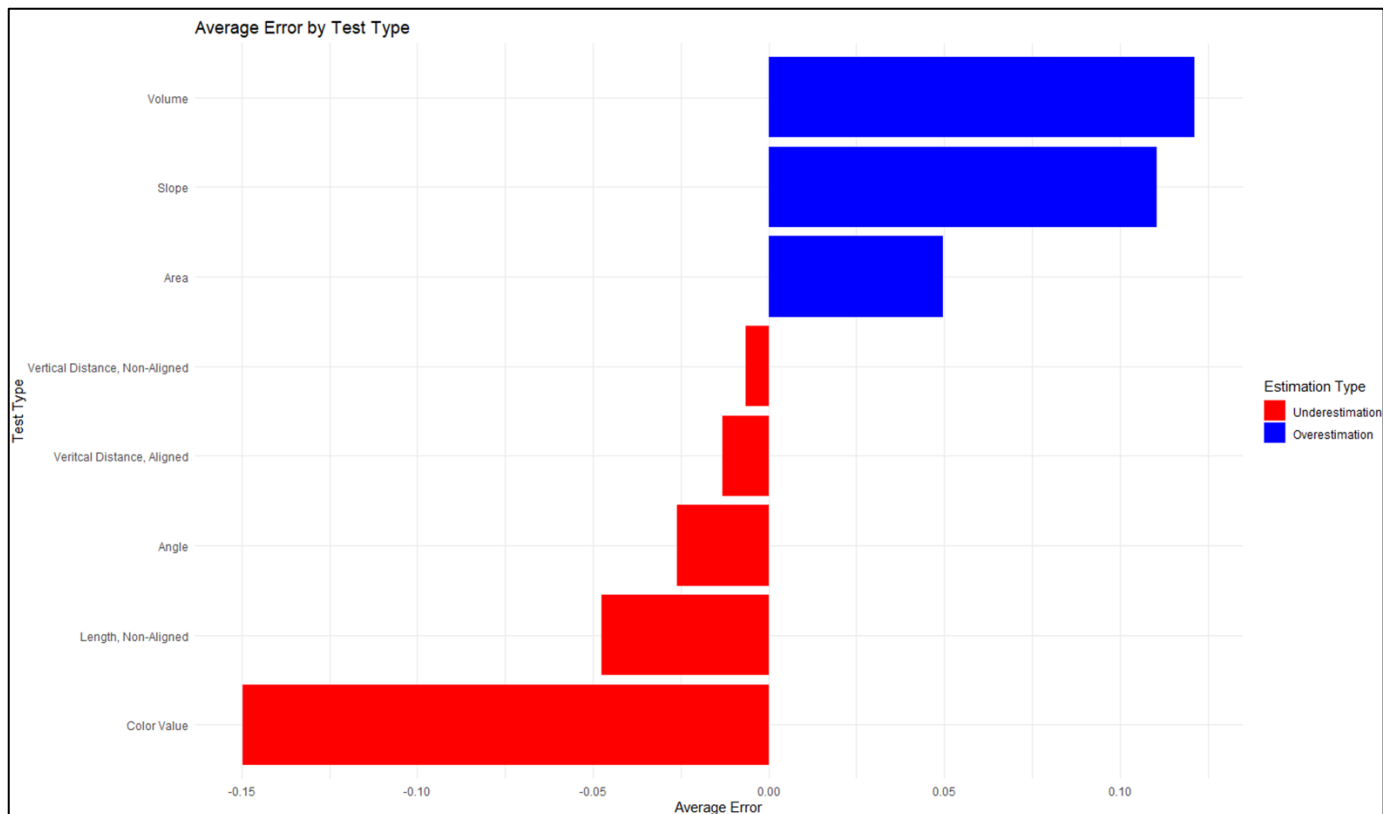
DSC 465

Homework 3

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1) a. Were there any tests where people generally underestimated or overestimated the data? Explain what field you can graph to test this, what graphical method reveals this clearly. Analyze the results and explain in a short paragraph.

```
> rm(list = ls())
> library(dplyr)
> library(ggplot2)
> data <- read.csv(file.choose(), header = TRUE)
> #create Error Column
> data$Error <- data$Response - data$TrueValue
> # Calculate Mean Error by Test
> test_error_summary <- data %>%
+   group_by(Test) %>%
+   summarise(MeanError = mean(Error)) %>%
+   arrange(MeanError)
> # Create the bar plot with two distinct colors for overestimation and underestimation
> ggplot(test_error_summary, aes(x = reorder(Test, MeanError), y = MeanError, fill = MeanError >
0)) +
+   geom_bar(stat = "identity") +
+   scale_fill_manual(values = c("red", "blue"), labels = c("Underestimation", "overestimation"))
+
+   coord_flip() +
+   labs(title = "Average Error by Test Type",
+         x = "Test Type",
+         y = "Average Error",
+         fill = "Estimation Type") +
+   theme_minimal()
```



The (graphical method) bar chart uses **red for underestimation (negative error)** and **blue for overestimation (positive error)**.

Observations:

Tests with Underestimation:

Color Value: The largest underestimation occurred here as shown by the long red bar extending to around -0.15. This suggests that participants significantly underestimated the values for this test. Other tests with noticeable underestimation include Length, Non-Aligned, Angle, Vertical Distance (Aligned and Non-Aligned), where the error is negative but smaller in magnitude.

Tests with Overestimation:

Volume: Participants generally overestimated this test, as seen by the large blue bar extending to around 0.12, indicating a positive average error. Other tests where overestimation occurred include Slope and Area, with participants consistently overestimating values in these tests.

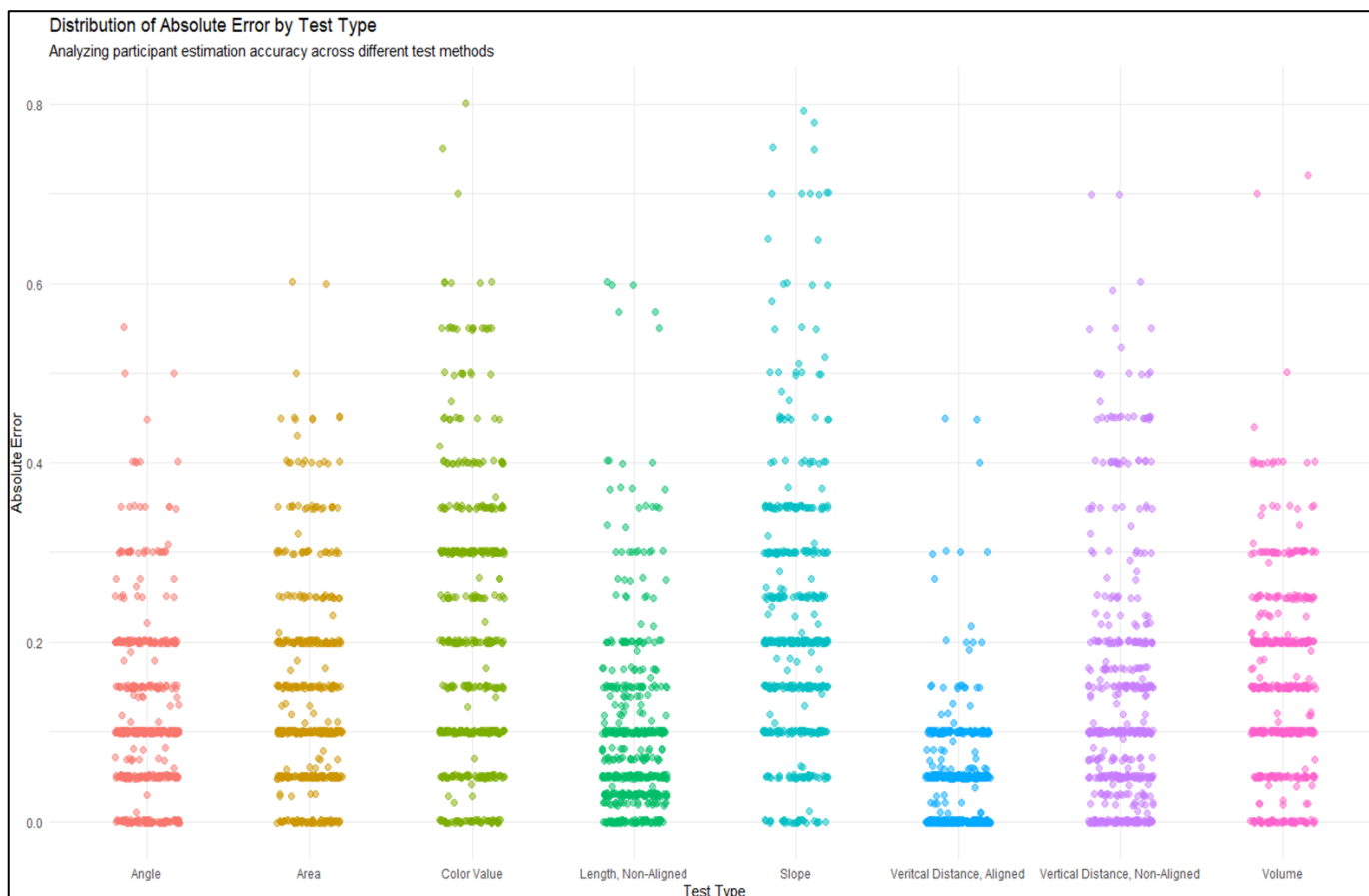
Conclusion:

The **Color Value test** had the most significant underestimation, with participants recording values much lower than the true value.

The **Volume test** had the most significant overestimation, with participants consistently recording values higher than the true value.

b. Use a univariate scatterplot or another technique that shows fine detail for a collection of distributions. For each Test (don't divide between Display 1 & 2 or Trial B, C and D) plot the AbsoluteError (absolute value of Error). Then write a short paragraph of analysis. How do the distributions of the data compare across the different methods our perception test studied for encoding numerical data visually? Is there any noticeable clumping of responses for any of the methods?

```
library(ggplot2)
# Create the AbsoluteError column
data$AbsoluteError <- abs(data$Error)
# Enhanced plot with jitter, color coding by Test, transparency, and improved labels
ggplot(data, aes(x = Test, y = AbsoluteError, color = Test)) +
  geom_jitter(width = 0.2, size = 2, alpha = 0.5) + # Adds jitter and transparency
  labs(
    title = "Distribution of Absolute Error by Test Type",
    subtitle = "Analyzing participant estimation accuracy across different test methods",
    x = "Test Type",
    y = "Absolute Error"
  ) +
  theme_minimal() + # Apply a clean theme
  theme(
    axis.text.x = element_text(angle = 0, hjust = 0.5), # Set x-axis labels to horizontal
    legend.position = "none" # Remove redundant legend
  )
```



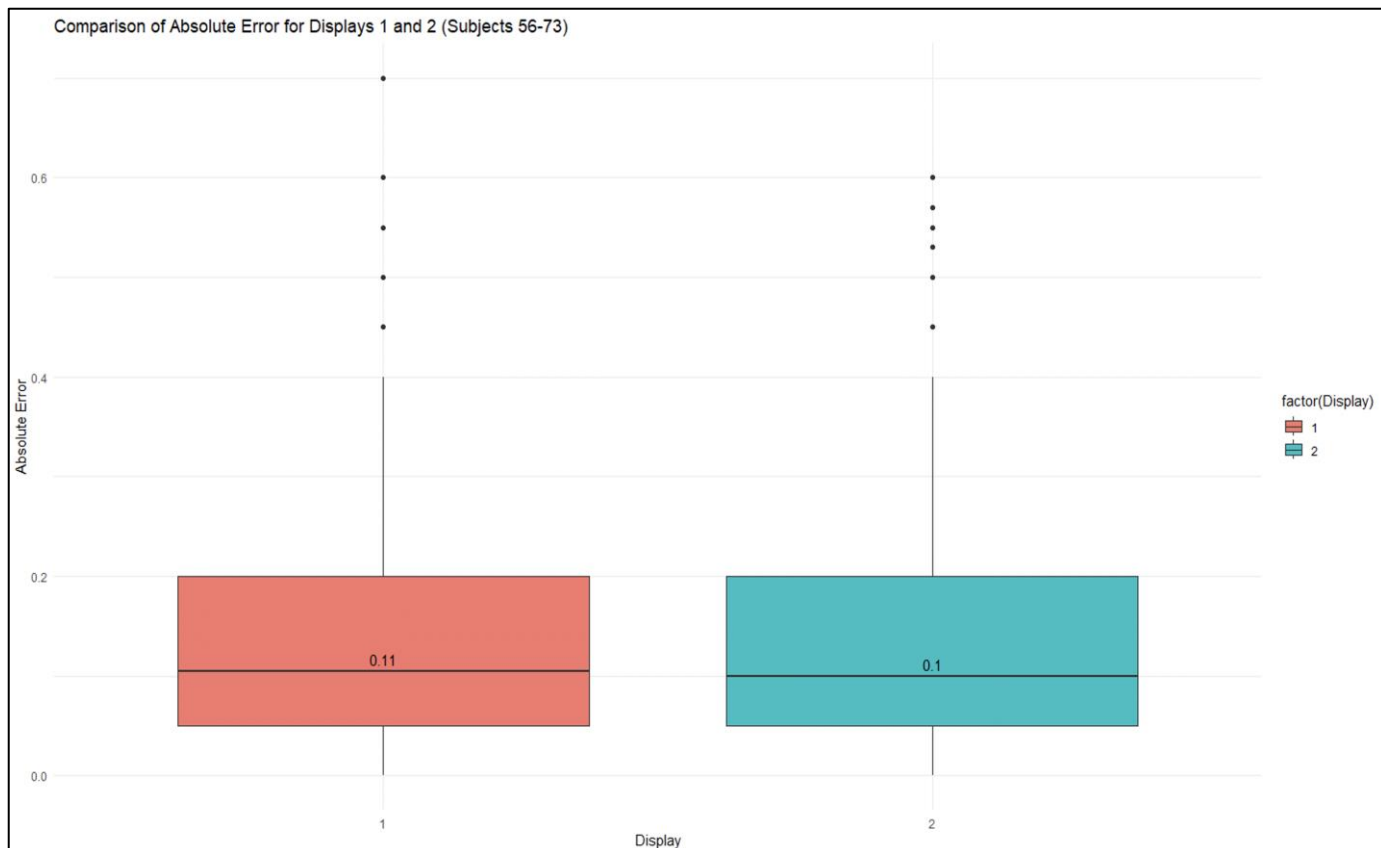
The **distribution of Absolute Error** varies significantly across the test types. Some test types, such as Color Value, Area and Vertical Distance (Aligned), show a more **compact**

distribution, indicating that the majority of participants had similar and consistent errors. This suggests that these encoding methods are easier for participants to interpret. On the other hand, test types like Volume, Slope and Length (Non-Aligned) exhibit a **wider distribution**, with errors spread across a larger range. This indicates more variability in the responses, suggesting that these encoding methods were harder for participants to interpret accurately.

There is **noticeable clumping of responses for certain methods**, particularly for **Color Value, Area, and Vertical Distance (Aligned)**. These methods show a tighter concentration of data points, meaning participants were more consistent in their responses for these visual encodings. Conversely, methods such as Volume and Slope show less clumping and more spread, indicating that participants had more difficulty estimating values consistently for these visual encodings.

c. Compare the data for Displays 1 and 2 for subjects 56-73 (you will need to filter the data in Tableau or R). Create a visualization that shows any differences in the response patterns between the two. These subjects all saw the first set of Displays before the second set. Is there any difference in the values for Displays 1 and 2? Did the participants get better at judging after having done it once?

```
> library(ggplot2)
> library(dplyr)
> # Assuming data has columns 'Response' and 'TrueValue', calculate AbsoluteError
> data$AbsoluteError <- abs(data$Response - data$TrueValue)
> # Filter data for Subjects 56 to 73
> filtered_data <- data %>%
+   filter(Subject >= 56 & Subject <= 73)
> # Plot the data
> ggplot(filtered_data, aes(x = factor(Display), y = AbsoluteError)) +
+   geom_boxplot(aes(fill = factor(Display))) +
+   stat_summary(fun = median, geom = "text", aes(label = round(after_stat(y), 2)),
+               position = position_dodge(width = 0.75), vjust = -0.5) + # Adding median labels
+   labs(title = "Comparison of Absolute Error for Displays 1 and 2 (Subjects 56-73)",
+        x = "Display",
+        y = "Absolute Error") +
+   theme_minimal()
```



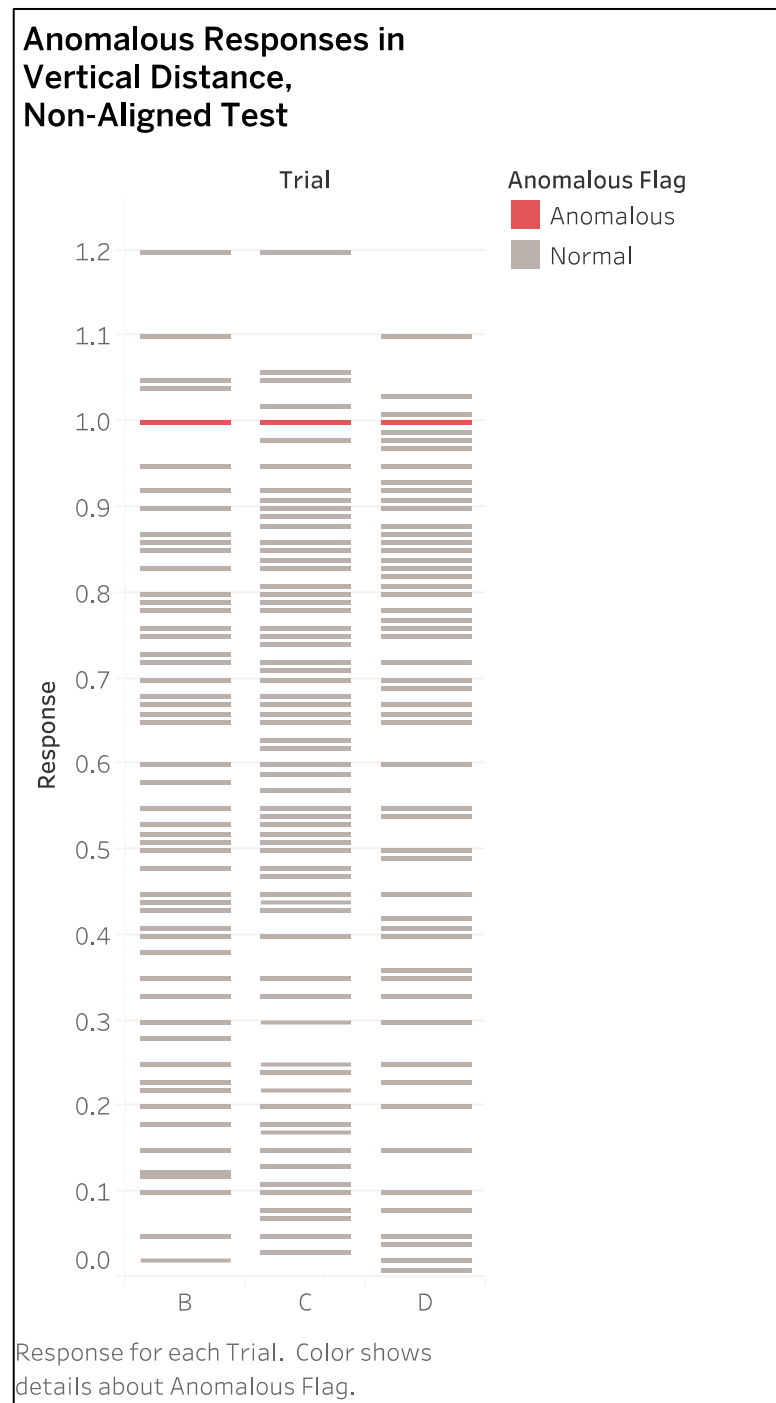
Median Comparison: The median Absolute Error for Display 1 is 0.11, while for Display 2 it is 0.10. This indicates that on average, participants' errors were slightly smaller in Display 2 than in Display 1. This small decrease in the median suggests that participants may have improved slightly after seeing the first display, but the difference is minimal.

Spread of Errors: The spread of errors (as represented by the size of the boxes) is similar between both displays, with Display 2 having a slightly tighter interquartile range (IQR), which means participants' responses were a bit more consistent in Display 2. Display 1 shows more extreme outliers, indicating that some participants made larger errors in the first display compared to the second.

While the median absolute error suggests a slight improvement in accuracy for Display 2, the overall spread of the errors is relatively similar with Display 1 showing more variability and larger outliers. This suggests that participants may have improved slightly after their first exposure to the displays but the improvement was not very significant for the group as a whole.

d. An erroneous stimulus was used for the first Display of "vertical distance, non-aligned" for a small subset of the subjects. They manifest themselves as an anomalous sequence of "1" Responses across Trial B, C and D. Look closely at the original raw scores and identify the sequence of subjects (hint: they are contiguous). Visualize the raw scores in a way that highlights these values and makes their anomalous nature clear. It should make it clear not only that they are outliers but should show any features that distinguish them from ordinary outliers. Some features that you might think about exploiting: they are identical values

across all three Trials, regardless of what the true values for the Trial is; they are only for a small subset of subjects.



The visualization identifies the anomalous sequence of "1" responses across Trials B, C, and D for the vertical distance, non-aligned test. These anomalous responses appear as a contiguous block of red lines, clearly distinguishable from the normal responses, which are gray. This subset of subjects consistently provided a response of "1" for all three trials, regardless of the actual trial conditions, making their responses stand out as distinct from other data points.

The clear separation of these responses highlights their uniformity and shows that the anomaly affects only a small subset of subjects. By using color to differentiate anomalous from normal responses, the plot emphasizes the repetition of "1" responses across trials. This makes it evident that the anomaly is not just an ordinary outlier but a systematic error affecting only a few individuals.

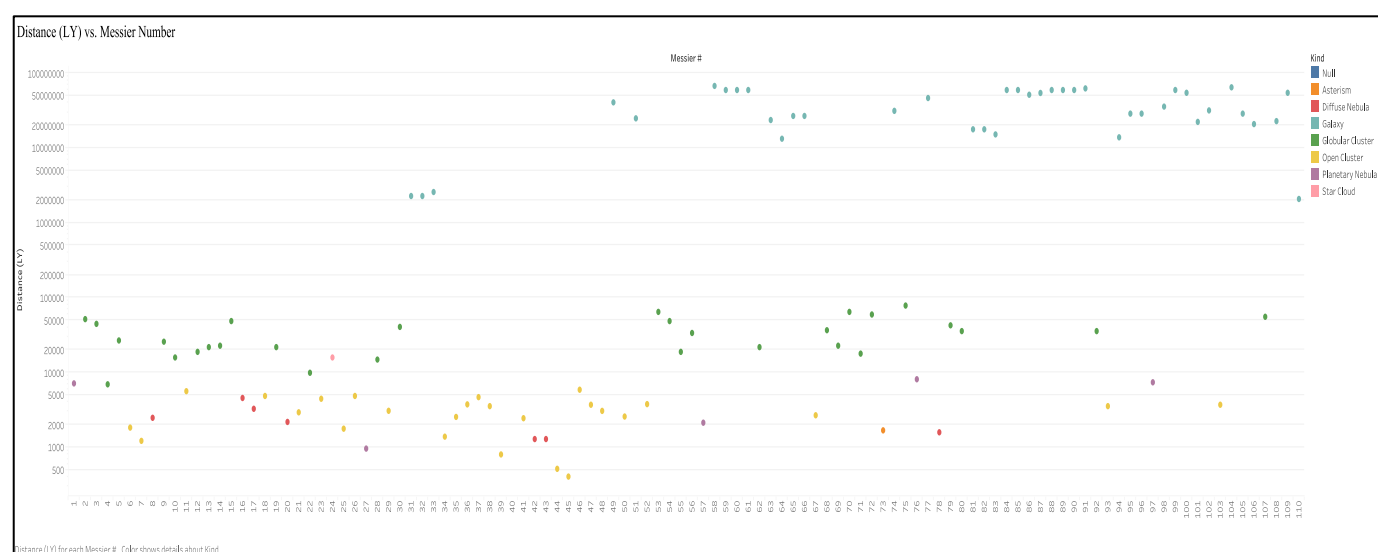
2) (20pts) Download the astronomical data for the Messier objects. These are objects that can be seen in a dark sky with binoculars or a telescope that Charles Messier cataloged in France in the 18th century so that they wouldn't be confused with comets. Some of these are clusters of stars or great clouds of gas in our galaxy, some are galaxies that are much farther away. The dataset contains a list of 100 deep sky objects along with their distances from the earth in light-years. Graph this data in the following ways to explore the information provided about these interesting objects.

For this dataset, you will have to pick suitable scales to make the data readable in your graphs. You should not wind up with a majority of the points squashed down along the one axis. In particular, for distances, the scale should show the "order-of-magnitude" of the distance in light years (10, 100, 1000, etc.) clearly.

a. Start by trying to graph one or more properties of the objects against the Messier Number. Remember, there is nothing 'intrinsic' about this number, it is just the order of Messier's list. Is there any property that exhibits a pattern with respect to the ordering in his list?

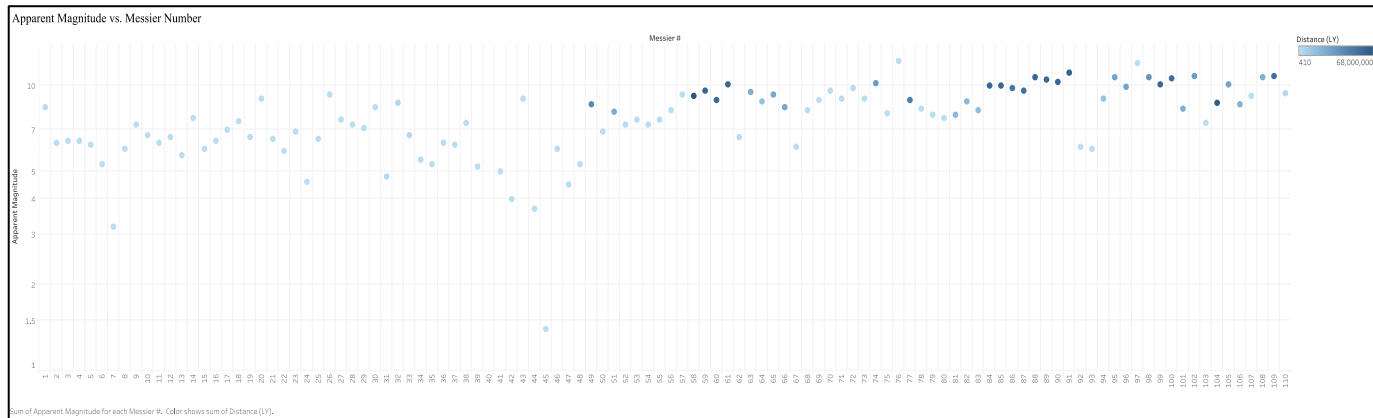
Ans: Since the dataset is large, the entire visualization is difficult to read on the document file, but I have still included the visualizations. After proper evaluation of the visualizations in the Tableau, we can say that no consistent pattern is observed between the Messier number and properties like distance, apparent magnitude, object type or size. The order of Messier's list does not seem to correlate with any of these properties, suggesting that the ordering of objects in the Messier catalog is arbitrary with respect to these characteristics.

Distance (LY) vs. Messier Number:



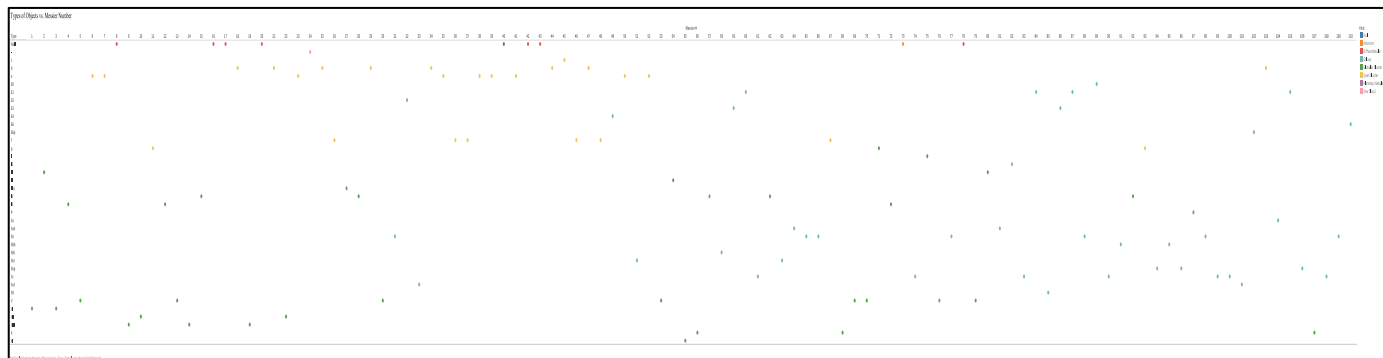
There does not appear to be a clear, consistent pattern between Messier number and distance. The distances are quite scattered across the range of Messier numbers, with some distant objects scattered among closer objects.

Apparent Magnitude vs. Messier Number:



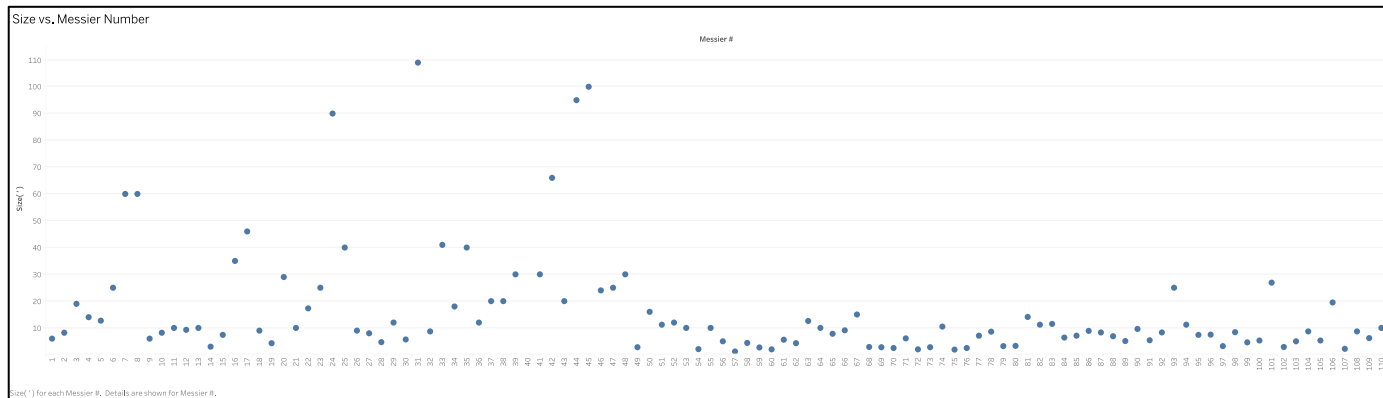
The apparent magnitude shows some clusters of values but also lacks a distinct pattern that corresponds with the order of Messier's list. The brightness of objects seems to be fairly spread across the list.

Type of Objects vs. Messier Number:



The types of objects seem randomly distributed across the Messier list, with no clear pattern of increasing or decreasing object types with the list order.

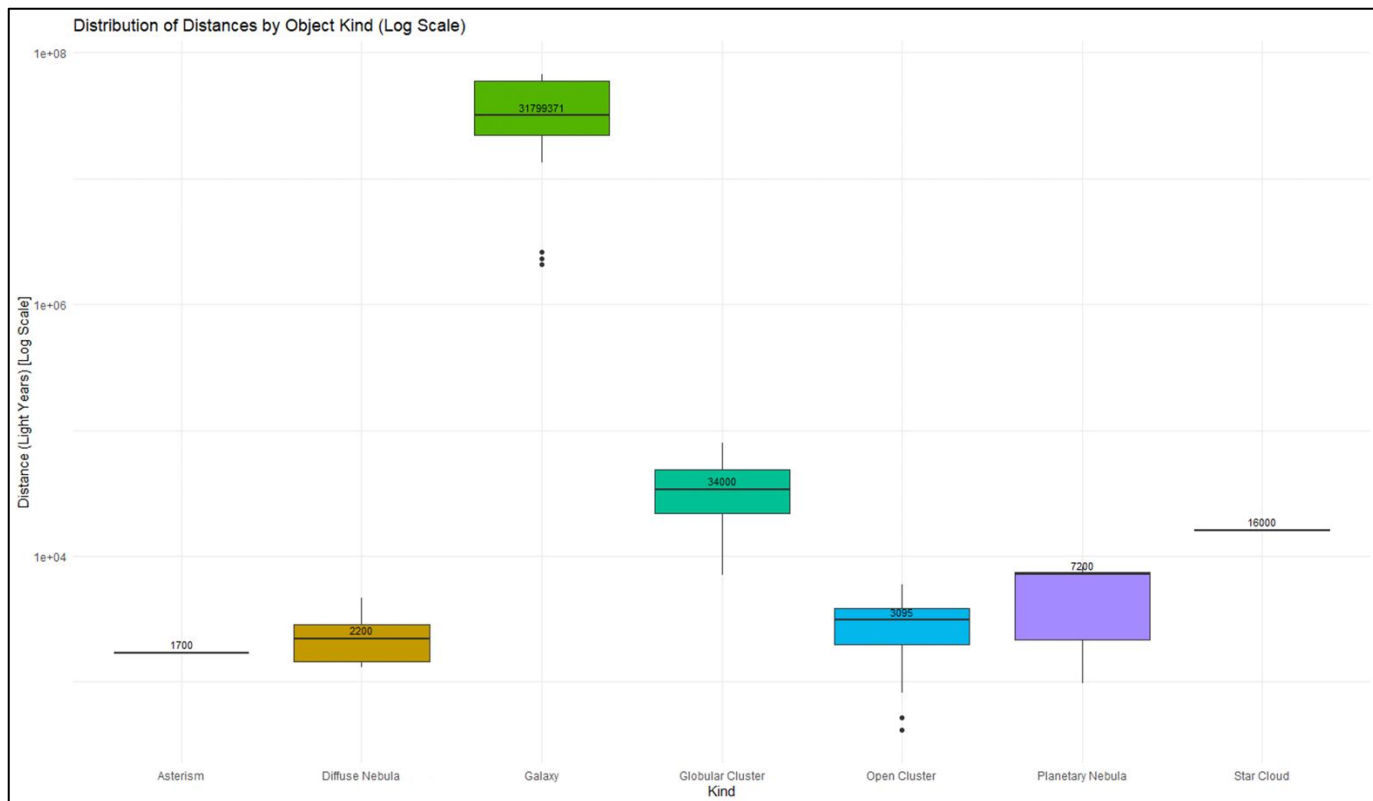
Size vs. Messier Number:



There is some variation in size across the Messier numbers, but there isn't a consistent increase or decrease in size values as the Messier number increases.

b. Create a visualization that compares the distributions of the distances to the objects in each Kind. Note that the Type variable is a very different category and is really a subcategory of Kind. Do not use that here. Sort the distribution displays in a way that makes the relationship clear.

```
> library(ggplot2)
> library(dplyr)
> messier_data <- read.csv(file.choose(), header = TRUE, fileEncoding = "latin1")
> # Clean the data by filtering out rows with missing or non-finite distance values
> messier_data_clean <- messier_data %>%
+   filter(is.finite(Distance..LY.)) %>%
+   filter(!is.na(kind))
> # Create a boxplot with median values using the updated syntax
> ggplot(messier_data_clean, aes(x = kind, y = Distance..LY., fill = kind)) +
+   geom_boxplot() +
+   scale_y_log10() + # Apply log scale to the y-axis
+   stat_summary(fun = median, geom = "text", aes(label = round(after_stat(y), 0)), vjust = -0.5,
size = 3, color = "black") +
+   labs(title = "Distribution of Distances by Object Kind (Log Scale)",
+         x = "Kind",
+         y = "Distance (Light Years) [Log Scale]") +
+   theme_minimal() +
+   theme(legend.position = "none") # Removes the legend if not needed
```



The boxplot effectively shows that different kinds of astronomical objects vary significantly in distance from Earth, highlighting the closer proximity of asterisms and diffuse nebulae compared to more distant star clouds.

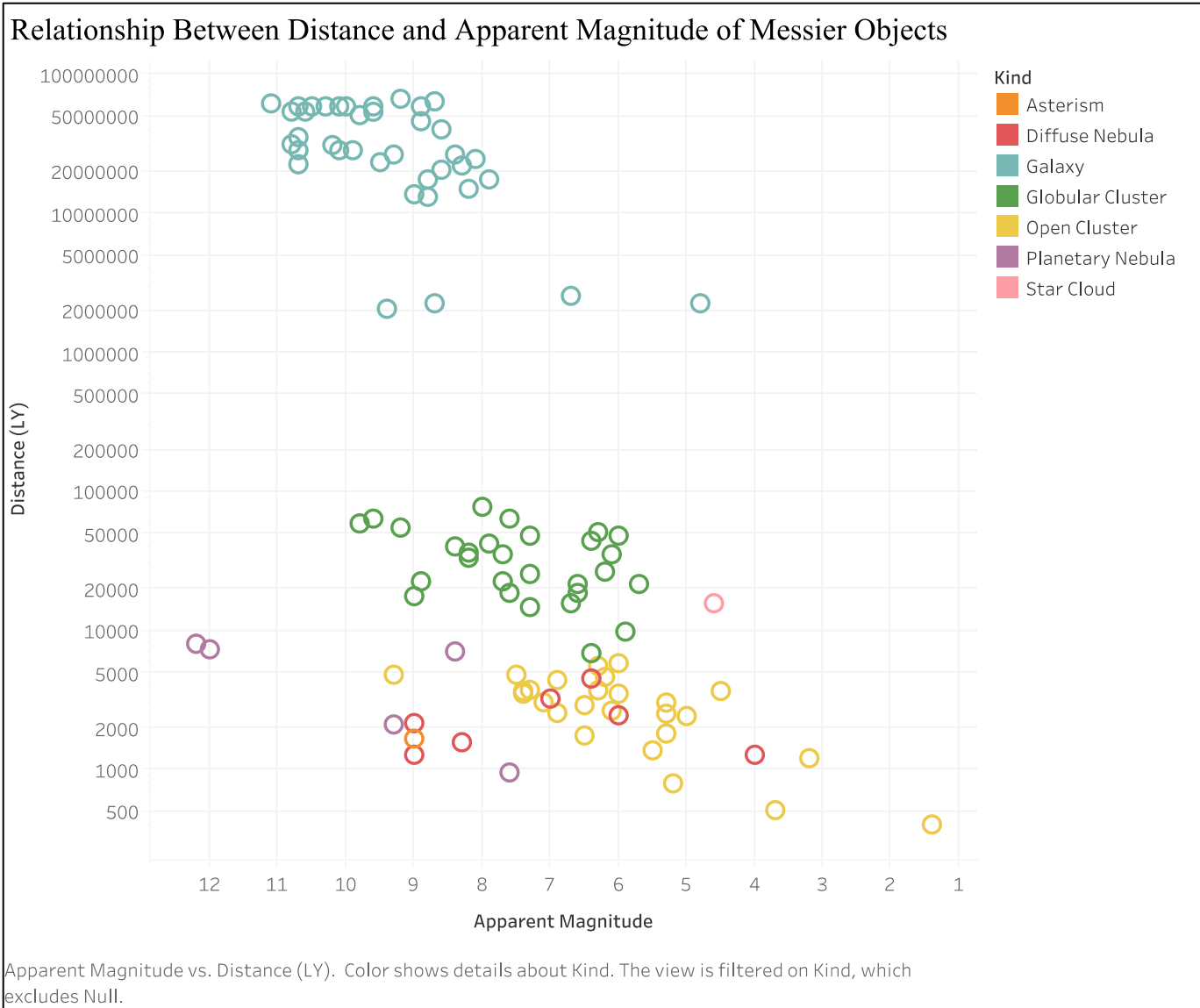
Asterisms and Diffuse Nebulae: These are the closest, with median distances of approximately 1,700 and 2,200 light years respectively.

Galaxies and Globular Clusters: Galaxies have a median distance of around 34,000 light years, while globular clusters are closer at about 3,095 light years.

Open Clusters and Planetary Nebulae: Open clusters are found at approximately 7,200 light years and planetary nebulae at about 16,000 light years.

Star Clouds: The furthest objects with a median distance of around 31,800,000 light years.

c. Create a scatter plot with the distance to the Messier objects plotted against their Apparent Magnitude (it's their visual magnitude, a measure of how bright they are in the sky). Note that these values may be... backwards from what you would think. The higher the number the fainter the object is in the sky. Try to incorporate that into your visualization to make the relationship clear.



Closer objects (those under 10,000 light-years) have a wide range of apparent magnitudes, suggesting that their intrinsic brightness (or their size and type) varies significantly. Farther objects (those over 100,000 light-years) tend to have higher apparent magnitudes (fainter), which is typical since light from distant objects gets more spread out. Different types of objects (e.g., galaxies, nebulae, clusters) are well spread across different distances and magnitudes, with some galaxies appearing much farther than other object types.

d. Augment the visualization in (c) by adjusting the size of the points in the scatter-plot based on the angular Size of the objects in the sky. Evaluate how easy it is to analyze all encoded aspects of the data from this graph and give a suggestion on how you might modify the graph to display all this information more readably.

Relationship Between Distance, Apparent Magnitude, and Angular Size of Messier Objects

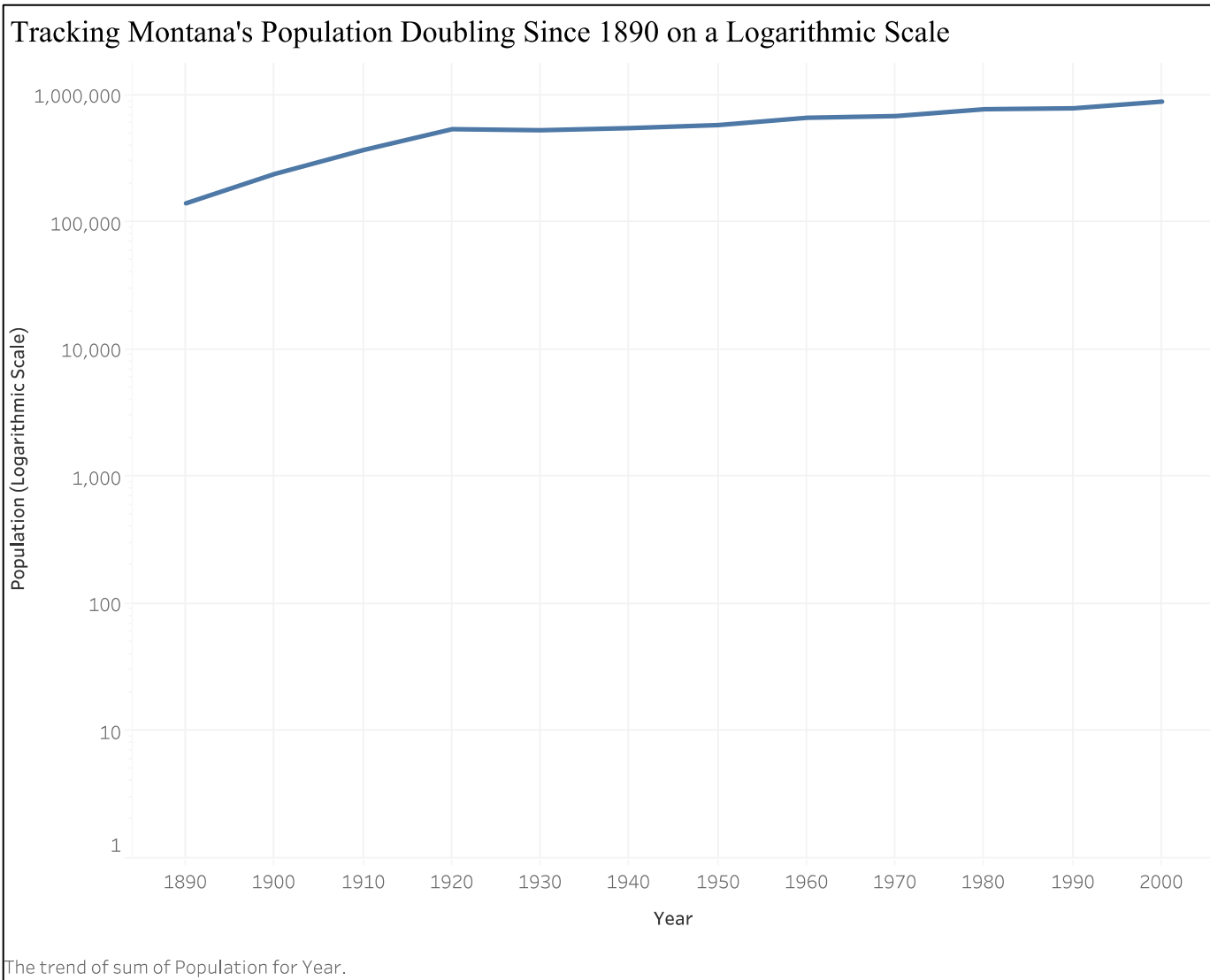


Apparent Magnitude vs. Distance (LY). Color shows details about Kind. Size shows details about Size('). The view is filtered on Kind, which excludes Null.

The scatter plot encodes three variables: distance, apparent magnitude and size, with color distinguishing the different types of objects. Larger circles represent objects with a larger angular size in the sky. Galaxies tend to have larger angular sizes and are farther away, as indicated by the size and position of the green and cyan points. The relationship between distance and apparent magnitude is still visible, but the addition of size makes the plot more complex.

3) (15pts) Download and graph the Montana Population data set (different from the one we used previously). Create visualizations using logarithmic scales, and intended for a technical audience, that clearly demonstrate visually the answers to the following questions. Viewers should be able to read the answers to these directly off the graph scales. Different logarithmic scale techniques may be appropriate for each part. If you use a single graph to answer multiple parts, make it clear that you are doing so.

a. How many times has the population doubled since 1890?



From what I can see on the graph:

In 1890, the population starts around 100,000.

By the 1950s, the population reaches 400,000.(approximately)

By 2000, the population is nearing 1,000,000.

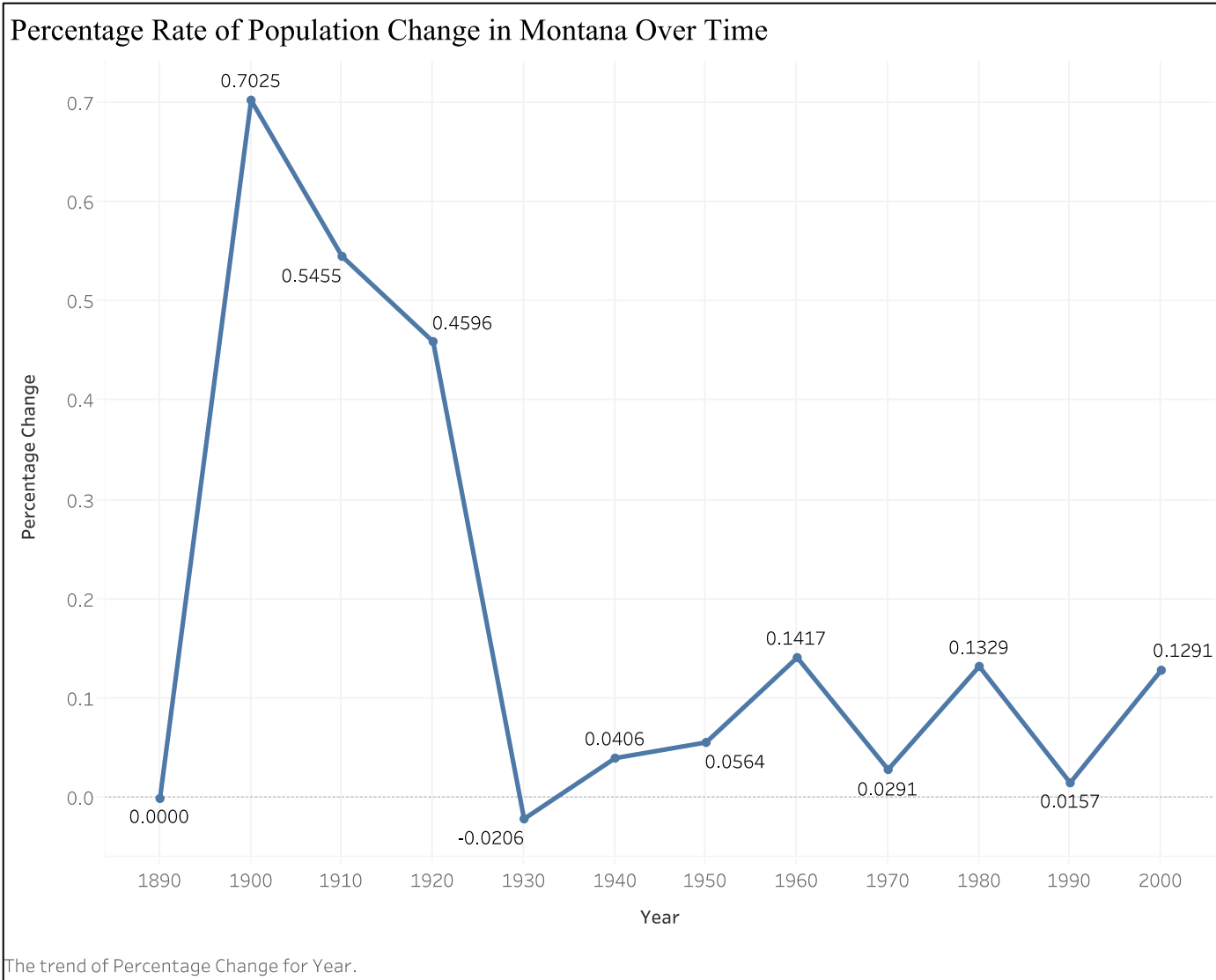
Thus, the population appears to have doubled approximately 3 times since 1890:

100,000 → 200,000 (1st doubling)

200,000 → 400,000 (2nd doubling)

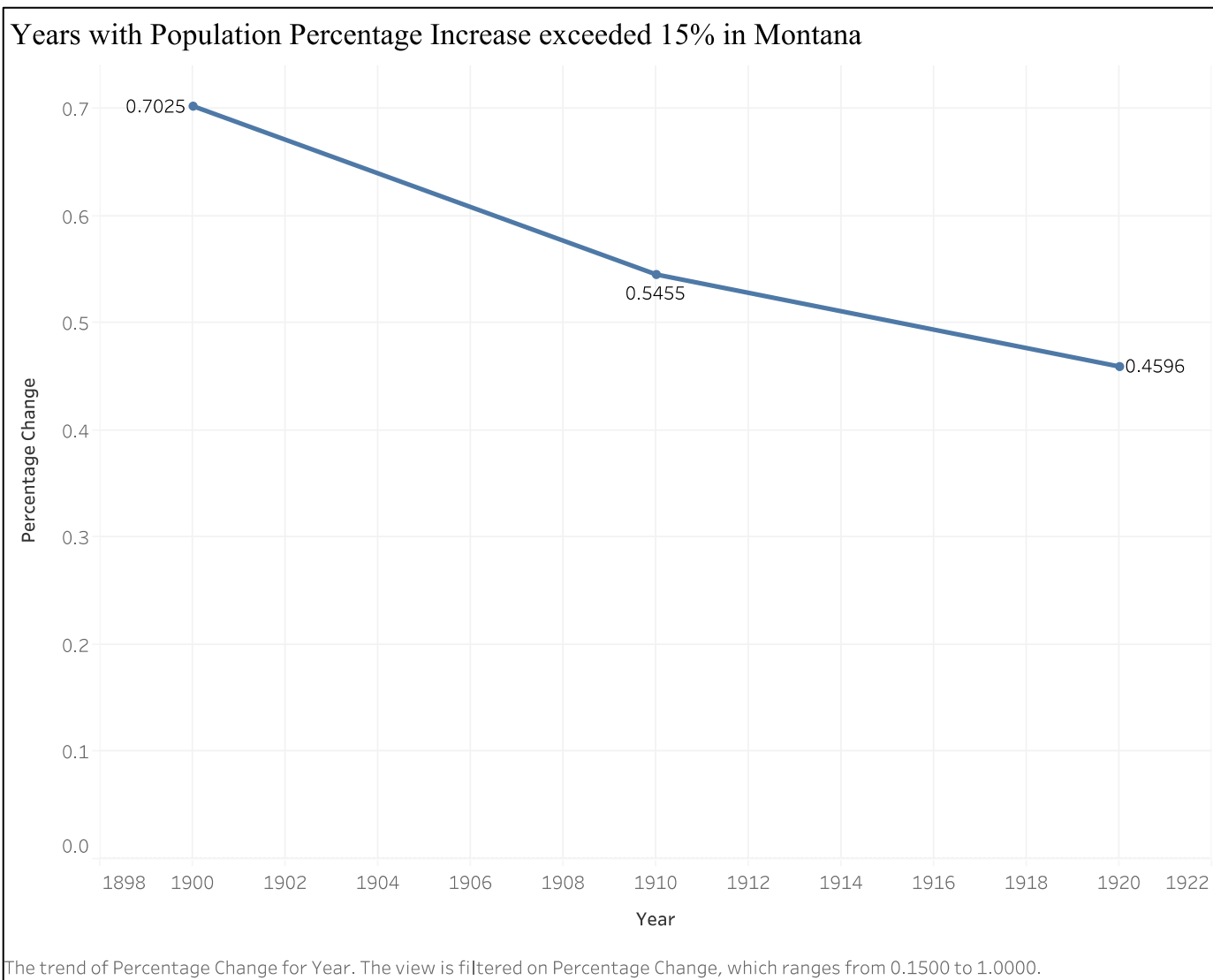
400,000 → 800,000 (3rd doubling)

- b. Has the percentage rate of change in the population increased or decreased over the years? What years had the greatest increase in population %-wise?



The percentage rate of change in Montana's population has decreased over the years, with the greatest increase occurring around 1900, when the population grew by 70.25%. After this peak, the rate generally declined, with the 1930s showing a slight negative change. There were smaller, fluctuating increases throughout the mid-20th century, but none reached the peak growth of the early 1900s.

- c. What years was the population percentage increase greater than 15%?



Based on the graph, the years where the population percentage increase was greater than 15% are: 1900, 1910, 1920.

These are the years where the percentage rate of change in the population exceeded 15%, as seen in your graph.

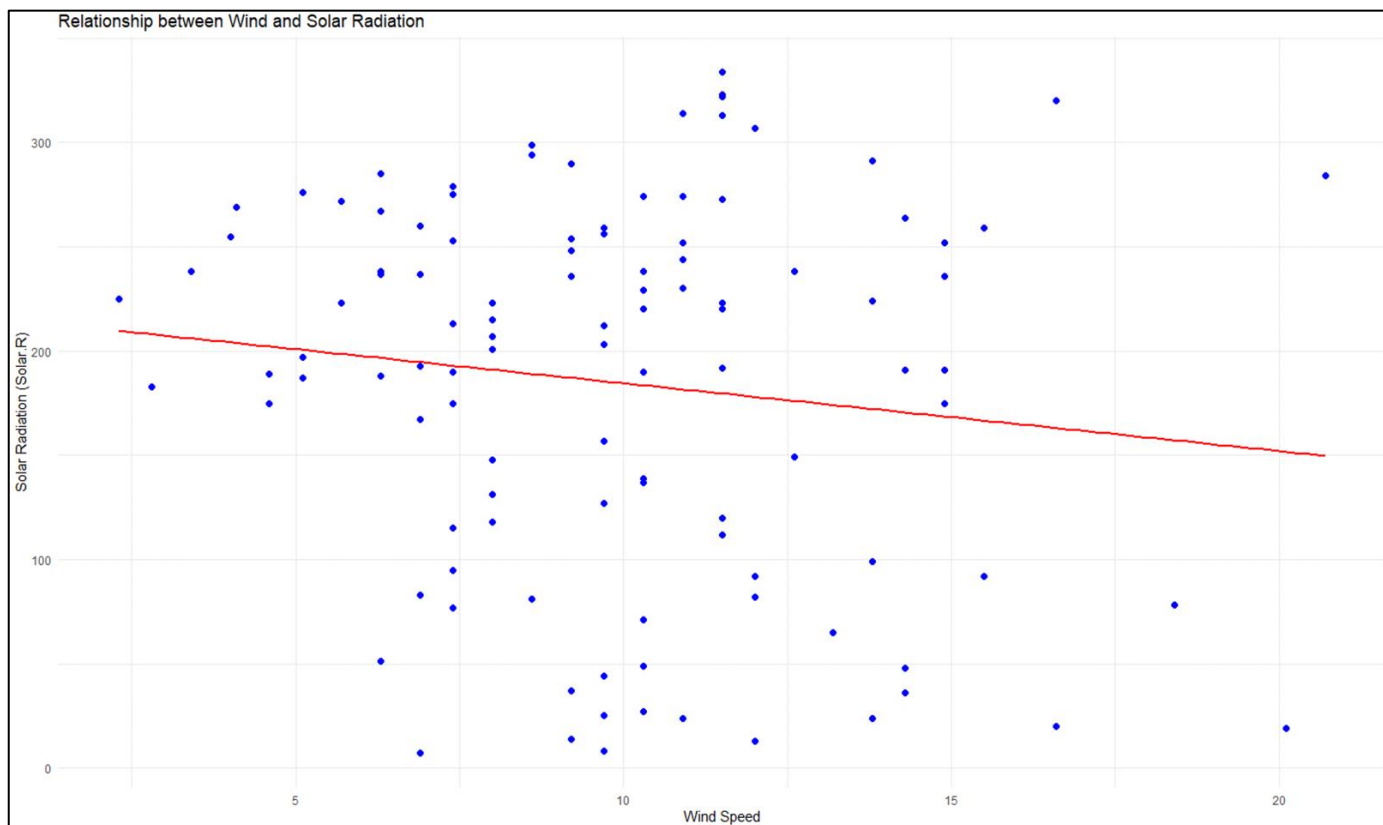
4) (20 pts) We will look at data on air quality, captured from May to September in New York. This is actually built into R, but not as a data frame. There is a copy on the D2L site.

a. Use a scatterplot to look at the relationship between Wind and Solar.R (solar radiation). Show a fit line. Make sure to produce a clean visualization with emphasis on the trend. This provides one view of the relationship.

```

> library(magrittr)
> library(ggplot2)
> airquality_data <- read.csv(file.choose(), header = TRUE)
> airquality_data <- na.omit(airquality_data)
> # Create a scatterplot of wind and Solar Radiation with a linear fit line
> ggplot(airquality_data, aes(x = wind, y = Solar.R)) +
+   geom_point(color = "blue", size = 2) + # Scatterplot points
+   geom_smooth(method = "lm", se = FALSE, color = "red") + # Add linear fit line
+   labs(title = "Relationship between wind and Solar Radiation",
+         x = "Wind Speed",
+         y = "Solar Radiation (Solar.R)") +
+   theme_minimal()
`geom_smooth()` using formula = 'y ~ x'

```



The scatterplot shows the individual data points (blue dots) and a linear regression line (red line) which is the fit line.

It suggests a slightly negative correlation between wind speed and solar radiation. As the wind speed increases, solar radiation tends to decrease slightly. The red fit line indicates this downward trend, but the spread of the data points shows that the relationship between wind speed and solar radiation is not very strong. The points are scattered around the line, indicating some variability, and the relationship is not perfectly linear.

b. Use a plot that will show the distributions of Wind and Solar.R and allow you to compare with fine detail.

Considering the `airquality_data` is already loaded.


```

> library(ggplot2)
> library(tidyr)

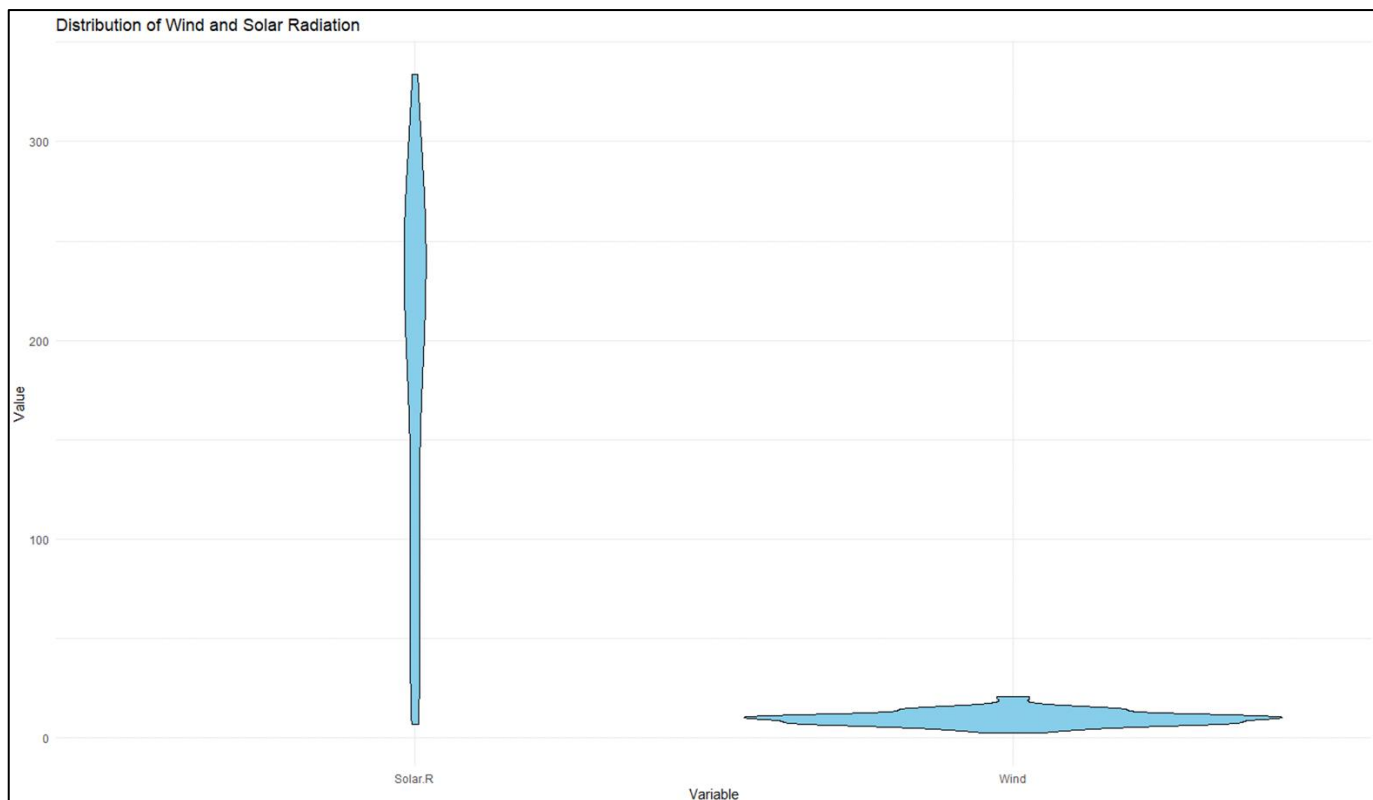
Attaching package: 'tidyr'

The following object is masked from 'package:magrittr':

    extract

> # Remove rows with missing values
> airquality_data <- na.omit(airquality_data)
> # Reshape the data to long format for comparison
> long_data <- pivot_longer(airquality_data, cols = c("wind", "solar.R"),
+                           names_to = "Variable", values_to = "value")
> # Create a violin plot to compare distributions of wind and solar.R
> ggplot(long_data, aes(x = variable, y = value)) +
+   geom_violin(fill = "skyblue", color = "black") +
+   labs(title = "Distribution of wind and solar Radiation",
+        x = "Variable",
+        y = "value") +
+   theme_minimal()

```



The distribution of solar radiation appears to be more tightly concentrated, as seen from the narrow width of the violin plot indicating that most values are clustered around a central value with fewer extreme values. The plot suggests a slightly right-skewed distribution with some values stretching towards higher levels of solar radiation.

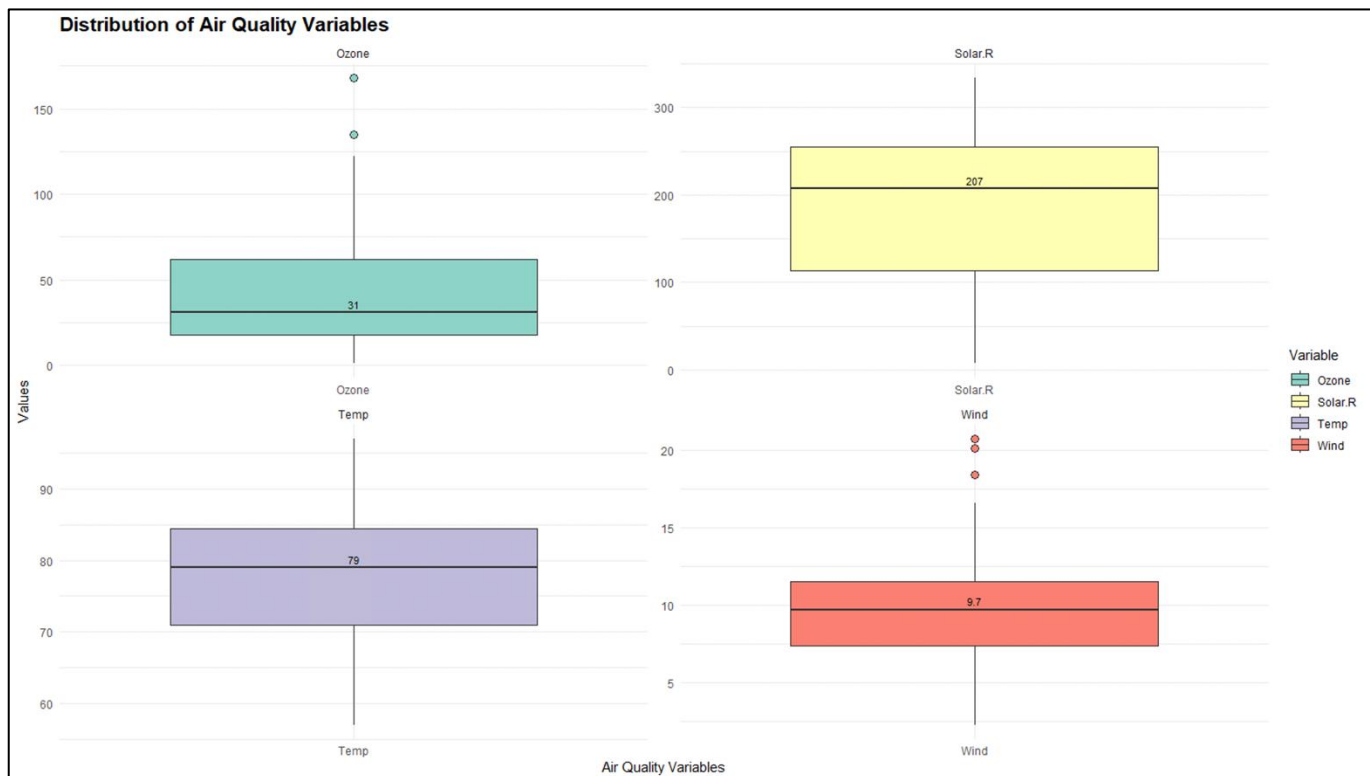
The wind speed distribution shows a broader spread, as indicated by the wider section of the violin plot in the middle. This suggests that wind speeds are more evenly distributed across a range of values with less extreme clustering compared to solar radiation. It also looks fairly symmetric.

c.Finally, show these distributions in context of the rest of the variables by using a

technique for comparing multiple distributions.

Note: you will need to transform the data in a particular way that we have studied. I showed in the Tableau tutorial and in an R tutorial. Hint – you need to collapse the current variables into two: (1) stores the original variable name, and (2) stores the corresponding original value.

```
> library(ggplot2)
> library(tidyr)
> # Remove rows with missing values
> airquality_data <- na.omit(airquality_data)
> # Reshape the data into long format for multiple variables
> long_data_all <- pivot_longer(airquality_data,
+                               cols = c(Ozone, solar.R, wind, Temp),
+                               names_to = "Variable",
+                               values_to = "Value")
> # Create boxplots to show distributions of all variables with facet wrap
> ggplot(long_data_all, aes(x = Variable, y = value, fill = Variable)) +
+   geom_boxplot(outlier.size = 3, outlier.shape = 21) + # Make outliers larger and distinct
+   stat_summary(fun = median, geom = "text", aes(label = round(.y.., 1)),
+               vjust = -0.5, size = 3, color = "black") + # Add median values
+   labs(title = "Distribution of Air Quality Variables",
+        x = "Air Quality Variables", # Descriptive x-axis label
+        y = "Values", # Descriptive y-axis label
+        caption = "Data from May to September in New York") + # Adding a caption
+   scale_fill_brewer(palette = "Set3", name = "Variable") + # Custom color palette and legend title
+   theme_minimal() +
+   theme(plot.title = element_text(face = "bold", size = 16)) + # Larger, bold title
+   facet_wrap(~ Variable, scales = "free")
```



The boxplot shows the distribution of four air quality variables: Ozone, Solar.R (Solar Radiation), Temp (Temperature) and Wind.

Ozone: The median value of Ozone is 31, which is labeled inside the box. The box (IQR) shows that most values of Ozone lie between approximately 20 and 70. There is one outlier above the upper whisker, indicating an unusually high Ozone value around 150.

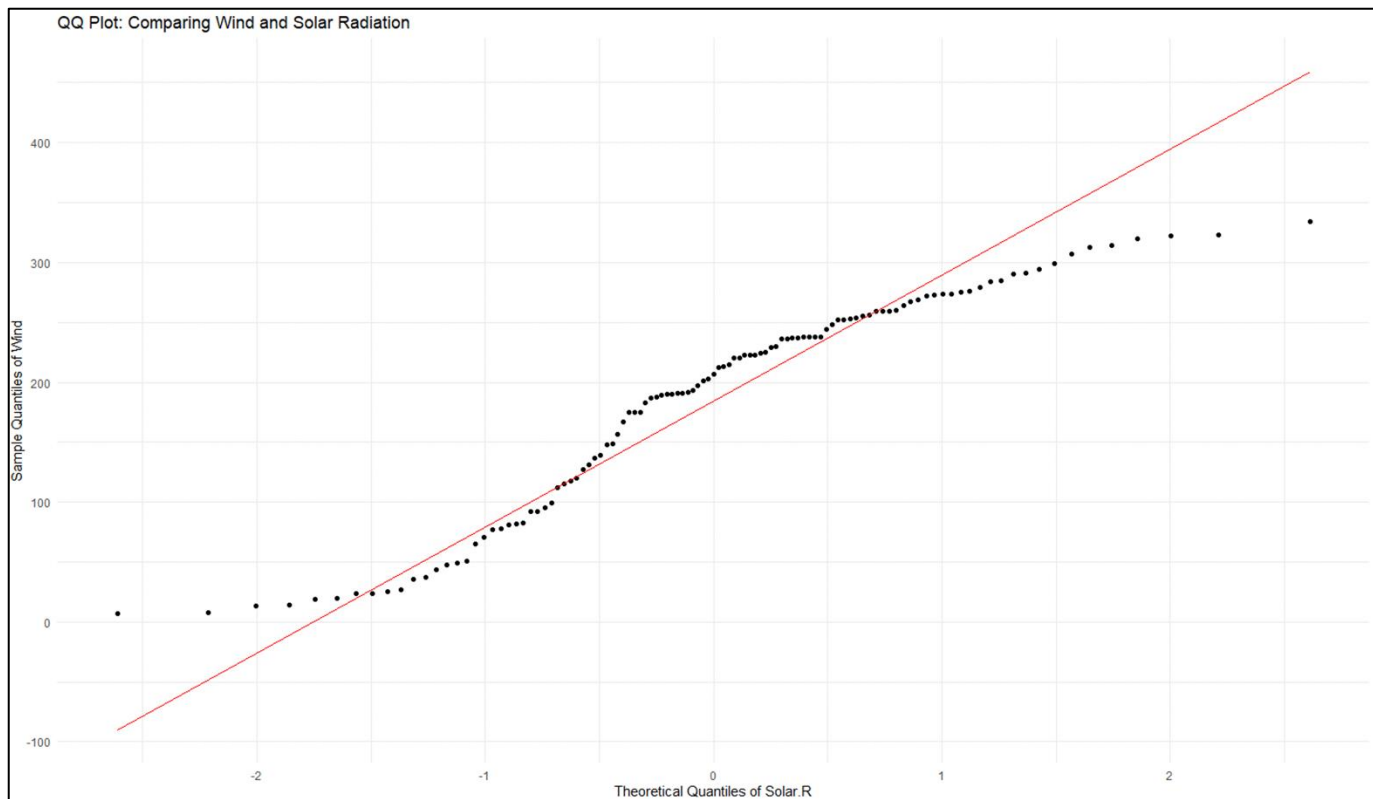
Solar Radiation (Solar.R): The median value of Solar.R is 207. The distribution of Solar Radiation is wider than the other variables, with values ranging from approximately 100 to 250. The distribution of Solar Radiation is fairly symmetric, as indicated by the equal length of whiskers above and below the box.

Temperature (Temp): The median value of Temperature is 79. The box shows that most temperature values lie between 70 and 85. The whiskers are quite narrow, indicating that the temperature values are tightly clustered around the median.

Wind: The median value of Wind speed is 9.7. Most wind speed values lie between 7 and 11. There are several outliers above the upper whisker, indicating unusually high wind speeds (above 15).

d. For extra credit, compare Wind and Solar.R again with a QQ plot. What does this tell you?

```
> library(ggplot2)
> # Remove rows with missing values
> airquality_data <- na.omit(airquality_data)
> # Create a QQ plot to compare wind and solar.R
> ggplot(data = airquality_data, aes(sample = wind)) +
+   stat_qq(aes(sample = Solar.R)) + # Compare wind and Solar.R
+   stat_qq_line(aes(sample = Solar.R), color = "red") + # Add a reference line
+   labs(title = "QQ Plot: Comparing wind and Solar Radiation",
+         x = "Theoretical quantiles of solar.R",
+         y = "Sample quantiles of wind") +
+   theme_minimal()
```



The red line in the Q-Q plot represents the reference line and the black points represent the quantiles of Wind plotted against the quantiles of Solar.R.

Near the reference line: When the points lie close to the red reference line, it indicates that the two variables (Wind and Solar.R) have similar distributions in those regions.

Deviations from the line: The points significantly deviate from the line, especially at the tails (both at the lower and upper quantiles) indicating that Wind and Solar.R have different distributions particularly in the extremes.